Graph Anomaly Detection via Multi-View Discriminative Awareness Learning

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***Abstract*—With the deeper research on attributed networks, graph anomaly detection is becoming an increasingly important topic. It aims to identify patterns deviating from a majority of nodes. Currently, graph anomaly detection algorithms based on reconstruction-based learning and contrastive-based learning have gained significant attention. To harness diverse supervised signals, an intuitive approach is to find an elegant strategy to fuse these two paradigms, forming the hybrid learning paradigm. Despite the success of the hybrid learning paradigm, due to its subgraph sampling based approach, it still grapples with issues related to unreliable neighborhood information and the neglect of topological details. To address these limitations, this paper proposes a new hybrid learning paradigm via multi-view discriminative awareness learning for graph anomaly detection. Unlike the previous hybrid learning paradigm, the graph reconstruction module fully incorporates attribute and topology information, enhancing the comprehensiveness of data reconstruction. Moreover, the multi-view discrimination module employs a view-level contrast method based on the complete graph, which helps to comprehensively extract the information in the attributed network and mitigates the neighborhood unreliability without increasing the complexity. The experimental results, obtained from a rigorous evaluation on six benchmark datasets, demonstrate the effectiveness of the proposed method compared to existing baseline methods.**

***Index Terms*—Attributed networks, graph anomaly detection, graph neural networks, self-supervised learning.**

I. INTRODUCTION

WITH the development of information technology, numerous fields have generated complex, interdependent, and interrelated data represented in the form of graphs [1], [2], [3]. These data are commonly known as attributed networks or attributed graphs. In attributed networks, nodes and edges represent multivariate information about entities and their complex relationships. With the extensive attention given to the study of attributed networks, notable progress has been made across various tasks within these networks, such as node classification, link prediction, and graph classification. Among these tasks, graph anomaly detection is a fundamental data mining task [4]. It has practical significance in various fields [5], including social spam detection [6], telecommunication fraud detection [7], network malware attack [8], and financial fraud detection [9].

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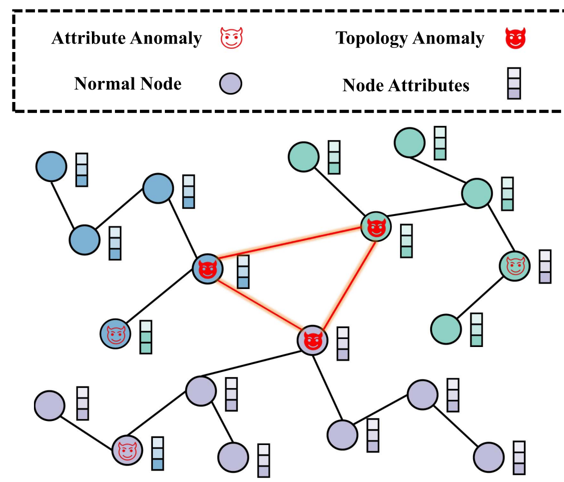


Fig. 1. An example illustrates two typical anomalies in the attributed networks: 1) Topology anomaly: incorrectly linked unrelated nodes; 2) Attribute anomaly: nodes with disturbed attributes. Nodes with the same color represent that they have similar attributes.

Graph anomaly detection aims to identify patterns that deviate from the majority of node specifications in the attributed networks [10]. Based on previous studies, anomalies can be categorized into two types: attribute anomaly and topology anomaly. As shown in Fig. 1, attribute anomaly occurs when node attributes are contaminated by factors like noise, while topology anomaly arises due to incorrect links between nodes. Despite the clear definition of graph anomalies, it is still a tough challenge to solve graph anomaly detection problems due to the complexity of graph data, the absence of labeled information, and the diversity of anomaly patterns.

Considering these challenges, extensive studies have been conducted in the domain of anomaly detection on attributed networks. Numerous traditional methods [11], [12], [13], [14], [15] have been widely explored. Nonetheless, limited to relying on domain-specific knowledge, they can not effectively capture deep nonlinear information. Fortunately, aided by deep learning,

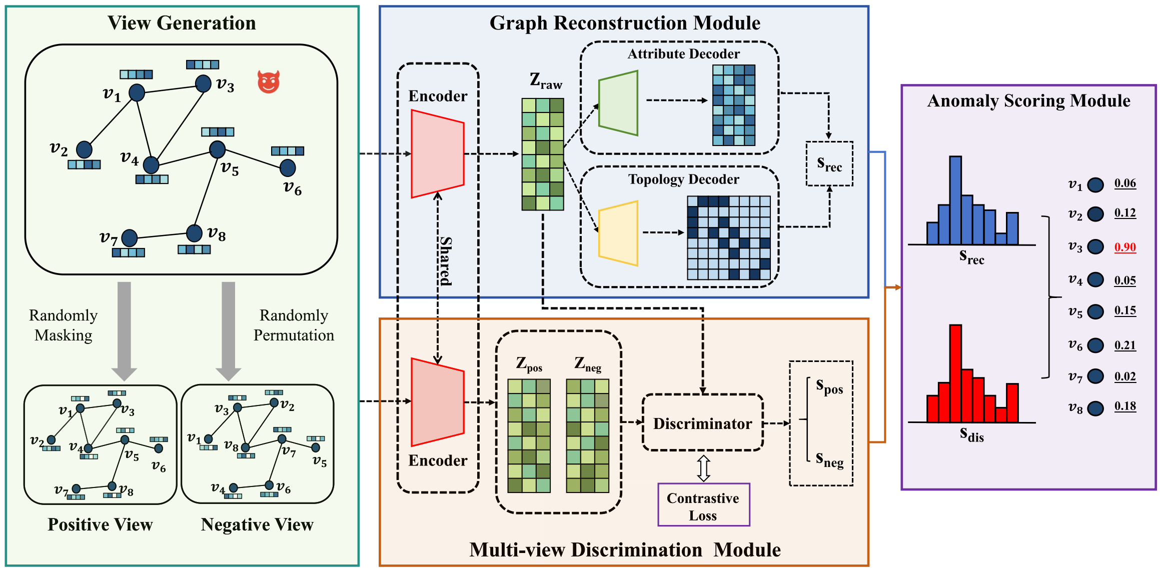


Fig. 2. The proposed framework consists of three parts: graph reconstruction module, multi-view discrimination module, and graph anomaly scoring module. Graph reconstruction module aims to identify anomalies by performing attribute regression and adjacency matrix regression on the original view through a graph encoding-decoding structure. Multi-view discrimination module aims to identify anomalies from potential patterns of anomalies by generating positive and negative views that are discriminated against the original view at the view level. Finally, the final anomaly score is computed through the graph anomaly scoring module.

recent advancements in graph anomaly detection have led to impressive results. Currently, graph anomaly detection models using reconstruction-based learning and contrastive-based learning have garnered considerable interest. In specific, the reconstruction-based learning paradigm [16], [17], [18] aims to measure the abnormality by calculating the reconstruction errors of the nodes using the graph autoencoder. Nevertheless, due to the sparseness of the anomaly samples, this approach is vulnerable to the class imbalance problem. In contrast, the contrastive-based learning paradigm [19], [20] utilizes a fixed-size subgraph sampling strategy for each node to design a proxy task that predicts the relationship between a node and a subgraph. This method utilizes node-subgraph pairing to establish an equal number of positive and negative instance pairs for node-level contrastive learning, effectively alleviating the class imbalance problem. However, fixed-size subgraph sampling may lead to the loss of neighborhood information of nodes with higher degrees, thus degrading the effectiveness of contrastive learning.

To leverage diverse supervised signals, prior studies [21], [22] have enhanced the contrastive-based learning paradigm by incorporating reconstruction learning, a method referred to as the hybrid learning paradigm. Nevertheless, by revisiting this paradigm, it is found that most methods still rely on fixed-size subgraph sampling, which focuses only on local information rather than global information. In particular, the incomplete capture of subgraphs and the presence of abnormal nodes result in the unreliability of neighborhoods. This undermines the completeness of reconstruction learning and fundamentally fails to address the limitations inherent in the contrastive-based learning paradigm. Additionally, the current hybrid learning paradigm only considers attribute reconstruction while ignoring topological information, making it difficult to identify topological anomalies.

To overcome the above weakness, in this paper, we propose a new hybrid paradigm via Multi-view Discriminative Awareness learning for Graph Anomaly Detection (MDA-GAD). The framework of the proposed method is shown in Fig. 2. Specifically, a graph anomaly detection model is designed, which includes a graph reconstruction module and a multi-view discrimination module that work in parallel. Unlike existing hybrid learning methods, which are limited to attribute reconstruction, the graph reconstruction module simultaneously reconstructs both attribute and topology, enhancing the ability to detect various anomalies. The multi-view discrimination module employs a sampling-free view generation method that generates positive and negative views directly from the original view for view-level discrimination learning. Finally, a graph anomaly scoring module is designed for the anomaly evaluation of each node. The proposed method utilizes complete graph information to ensure a comprehensive representation on the attributed networks, effectively reducing the neighborhood unreliability without increasing complexity. Overall, the main contributions are summarised as follows:

- Present a novel hybrid learning approach that effectively integrates contrastive-based and reconstruction-based learning paradigms for anomaly detection on attributed networks.
- Introduce a new method to generate views and shift the focus from node-level to view-level contrast, overcoming the drawbacks of previous hybrid learning methods regarding neighborhood unreliability and high complexity.
- The experimental results illustrate the effectiveness of the proposed method, showing its superiority over existing baselines and its high time efficiency.

The rest parts of this paper are organized as follows. In Section II, most relative approaches to the proposed method are introduced. Then, problem definitions and notations are introduced in Section III. Section IV discusses the proposed framework in detail, and comprehensive experiments are conducted in Section V. Finally, Section VI presents the conclusions of this work.

II. Related Work

In this section, we provide a brief overview of related works, encompassing graph anomaly detection and graph self-supervised learning.

A. Graph Anomaly Detection

As the rise of attributed networks, graph anomaly detection algorithm has garnered attention from many researchers. Numerous machine learning algorithms have been widely explored in the field of anomaly detection. Perozzi et al. [12] uses the ego-network information of nodes to detect anomalous neighbors. Li et al. [14] analyzes the attribute information residuals and their consistent representation. Further, Peng et al. [15] combines residual analysis and CUR decomposition for anomaly detection. All of the mentioned methods demonstrate excellent performance in low-dimensional attributed networks. Nonetheless, due to their shallow learning mechanism, these algorithms may struggle to effectively detect anomalies with high dimensions.

In recent years, deep learning-based anomaly detection algorithms have been increasingly explored with impressive achievements. Ding et al. [16] employs a graph convolutional network (GCN) [23] based autoencoder to quantify the anomaly degree of a node by calculating the reconstruction loss of both the adjacency matrix and the attribute matrix. Fan et al. [24] integrates graph attention networks (GAT) [25] into the autoencoder framework. Pei et al. [17] improves GAT by adding residual information. Luo et al. [18] offers a community-aware tailored GCN to mitigate the over-smoothing of anomalous node representations. Roy et al. [26] develops a method by reconstructing the self-feature, degree, and neighborhood node distribution representations. Nevertheless, the reconstruction-based learning paradigm is susceptible to class imbalance. To overcome this limitation, Liu et al. [19] first employs a contrastive-based learning approach that utilizes metric instance pairs to achieve node-subgraph contrast. Jin et al. [20] and Duan et al. [27] enhance the contrastive-based learning paradigm by incorporating node-node contrast and subgraph-subgraph contrast into node-subgraph contrast gradually, enabling anomaly detection through multi-scale contrastive learning. Duan et al. [28] introduces normality learning in the contrastive-based learning paradigm to mitigate the damage of abnormal nodes. Pan et al. [29] designs a ego-neighbor comparison learning method to efficiently detect graph anomalies. It is noteworthy that the aforementioned methods are confined to the contrastive-based learning paradigm and might not fully exploit the model's learning capacity with respect to supervised signals. Zheng et al. [21] proposes a hybrid learning paradigm from the joint perspective of the contrastive-based learning paradigm and the reconstruction-based learning paradigm. Zhang et al. [22] improves the hybrid learning approach by utilizing a multi-view mechanism that includes both original and diffuse perspectives. Nonetheless, these methods still rely on fixed-size subgraph sampling and focus solely on attribute reconstruction, neglecting the combined reconstruction of both attribute and topology. To address this issue, the proposed method introduces a sampling-free contrastive strategy and considers the integrity of the reconstruction.

B. Graph Self-Supervised Learning

The emergence of self-supervised learning brings a new paradigm for graph learning. It focuses on extracting graph representations by designing suitable tasks that no longer require manually labeled data. Generation-based models and contrast-based models are the two dominant graph self-supervised types [30].

Generation-based models use a graph autoencoder to obtain a representation by reconstructing graph information. The earliest models, including GAE and VGAE [31], concentrate on reconstructing the adjacency matrix to acquire graph embeddings. Building upon the effective masking strategy in computer vision [32], Tan et al. [33] improves GAE by incorporating edge masking. Furthermore, Li et al. [34] introduces a novel path masking strategy that enhances the graph mask autoencoder. Hou et al. [35] adapts the graph mask autoencoder by utilizing a double-masking strategy and scaled cosine error. Wang et al. [36] introduces an easy-to-hard adversarial masking strategy to enhance alignment performance by generating challenging samples.

Contrast-based models are designed with suitable pretext tasks that maximize the mutual information of positive samples. Velivckovic et al. [37] pioneers graph self-supervised contrastive learning by maximizing mutual information at both node and graph levels to capture the global structure. On this basis, Hassani et al. [38] proposes contrastive learning through original and diffuse graphs. Lin et al. [39] proposes a spectral augmentation scheme to generate topological augmentation through perturbation mapping to improve the performance. Shen et al. [40] utilizes a multi-head attention mechanism to learn graph augmented views with adaptive topology for conrtastive learning.

These works have shown desirable performance in downstream tasks, such as node classification and link prediction. Nevertheless, there have been very limited attempts for anomaly detection tasks. Motivated by the above methods, an anomaly detection framework is designed from both generative and contrastive perspectives.

III. Problem Definition

In this section, we define the problems associated with node anomaly detection on attributed networks. For better understanding, Table I presents the notations along with their descriptions. Accordingly, the definition of attributed networks is given as below.

For a given attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ and $\mathcal{E}$ represent the set of nodes and edges, which can be formalized by the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{X} \in \mathbb{R}^{N\times F}$ denotes the

TABLE I
NOTATIONS AND DESCRIPTIONS

Notation	Descriptions
$\mathcal{G}=(\mathcal{V},\mathcal{E},\mathbf{X})$	The attributed network
$\mathcal{V},\mathcal{E}$	The node and edge set of $\mathcal{G}$
$\mathbf{A}\in\mathbb{R}^{N\times N}$	The adjacency matrix of $\mathcal{G}$
$\mathbf{X}_{raw}\in\mathbb{R}^{N\times F}$	The node feature matrix of $\mathcal{G}$
$\mathbf{Z}_{raw}\in\mathbb{R}^{N\times D}$	The node embedding matrix for $\mathbf{X}_{raw}$
$\widehat{\mathbf{A}}\in\mathbb{R}^{N\times N}$	The reconstructed adjacency matrix
$\widehat{\mathbf{X}}_{raw}\in\mathbb{R}^{N\times F}$	The reconstructed attribute matrix
$\mathbf{W}_{enc}\in\mathbb{R}^{D\times D}$	The weight matrix of the encoder
$\mathbf{W}_{Adec}\in\mathbb{R}^{D\times D}$	The weight matrix of the attribute decoder
$\mathbf{W}_{Tdec}\in\mathbb{R}^{D\times D}$	The weight matrix of the topology decoder
$\mathbf{X}_{pos}\in\mathbb{R}^{N\times F}$	The node feature of the positive view
$\mathbf{X}_{neg}\in\mathbb{R}^{N\times F}$	The node feature of the negative view
$\mathbf{Z}_{pos}\in\mathbb{R}^{N\times D}$	The node embedding of the positive view
$\mathbf{Z}_{neg}\in\mathbb{R}^{N\times D}$	The node embedding of the negative view
$\mathbf{s}_{pos}\in\mathbb{R}^{N\times 1}$	The similarity scores of $\mathbf{Z}_{raw}$ and $\mathbf{Z}_{pos}$
$\mathbf{s}_{neg}\in\mathbb{R}^{N\times 1}$	The similarity scores of $\mathbf{Z}_{raw}$ and $\mathbf{Z}_{neg}$
$\mathbf{W}_{dis}\in\mathbb{R}^{D\times D}$	The weight matrix of the discriminator
R	The rounds for anomaly score calculation
N	The number of nodes in $\mathcal{G}$
F	The dimension of node features in $\mathcal{G}$
D	The dimension of node embedding matrix
$\text{score}(v_i)$	The anomaly score of v_i

attribute matrix of all nodes. Thus, the attributed network is also expressed as $\mathcal{G}=(\mathbf{A},\mathbf{X})$.

The focus of this paper is the unsupervised node-level anomaly detection problem on attributed networks, which is formalized as follows:

For a given attributed network $\mathcal{G}=(\mathcal{V},\mathcal{E},\mathbf{X})$, the objective of this task is to compute a score and rank the abnormality degree of each node v_i by defining a valid anomaly function, which can be formalized as $g(\cdot):\mathbb{R}^{N\times F}\rightarrow\mathbb{R}^{N\times 1}$. Typically, nodes with high scores are considered as abnormal nodes.

IV. The Proposed Method

In this section, we outline the overall framework of the algorithm for detecting node-level graph anomalies in a self-supervised manner. The proposed method has three distinct components, including graph reconstruction module, multi-view discrimination module, and graph anomaly scoring module. First, the graph reconstruction module is designed to identify node anomalies by measuring the degree of mismatch between the reconstructed and the original views using the regression loss generated by the reconstruction. Second, the multi-view discrimination module is designed to generate high-quality positive and negative views to detect anomalies through a view-level contrastive learning task. Finally, the anomaly score for all nodes is computed by aggregating the scores from the first two modules. Multiple rounds of anomaly scores are averaged to mitigate the randomness associated with view generation.

A. Graph Reconstruction Module

Abnormal nodes exhibit patterns that differ from the typical behaviors observed in the majority of normal nodes. Among existing anomaly detection methods, deep autoencoders are considered to have strong potential. The degree of node abnormality can be quantified by the reconstruction loss. To reconstruct the graph information efficiently, a graph autoencoder structure is employed. Empowered by the autoencoder, the proposed framework effectively captures and reproduces the complex relationships within the graph, facilitating the identification of anomalies through disparities in both topology and attribute information.

View Encoder: A generalized approach is adopted to encode the original view by GCN. Utilizing the inherent efficient modeling and information aggregation capabilities of GCN, the encoder adeptly acquires a comprehensive representation of the network. The representation of each graph convolutional layer is formulated as follows:

$$\mathbf{H}^{(\ell+1)}=\sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(\ell)}\mathbf{W}^{(\ell)}\right), \tag{1}$$

where $\mathbf{H}^{(\ell)}$ and $\mathbf{H}^{(\ell+1)}$ are the inputs and outputs of the ℓ-th convolutional layer respectively, $\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$ denotes a symmetric normalization of the adjacency matrix, $\tilde{\mathbf{A}}$ represents the adjacency matrix with self-loop, and $\mathbf{W}^{(\ell)}$ represents the shared weight matrix of all nodes in the network. $\sigma(\cdot)$ represents a non-linear activation function. Then, the view encoder is defined as:

$$\mathbf{Z}_{raw}=\text{GCN}_{enc}\left(\mathbf{X}_{raw},\mathbf{A};\mathbf{W}_{enc}\right), \tag{2}$$

where $\mathbf{Z}_{raw}$, $\mathbf{X}_{raw}$, $\mathbf{A}$, $\mathbf{W}_{enc}$ denote the node embedding matrix, feature matrix, the adjacency matrix, and the weight matrix of the encoder, respectively.

Topology Reconstruction Decoder: The topology reconstruction decoder aims to restore node structure information and can effectively detect structural anomalies in the graph. The GCN acts as a topology decoder, reconstructing node topology information from latent node representations. The topology reconstruction decoder can be represented as:

$$\mathbf{H}=\text{GCN}_{Tdec}\left(\mathbf{Z}_{raw},\mathbf{A};\mathbf{W}_{Tdec}\right), \tag{3}$$

$$\widehat{\mathbf{A}}=\mathbf{H}\mathbf{H}^{T}, \tag{4}$$

where $\mathbf{H}$, $\mathbf{W}_{Tdec}$ denote the potential node embedding matrix after the decoder and the weight matrix of the topology decoder. $\widehat{\mathbf{A}}$ denotes the reconstructed adjacency matrix obtained from the inner product of $\mathbf{H}$.

Attribute Reconstruction Decoder: The attribute reconstruction decoder is designed to regress node attributes and serves as a powerful tool for detecting contextual anomalies on graphs. Similarly, the GCN is employed as an attribute decoder. If the node attributes can be effectively approximated using the decoder, which suggests that the node is less likely to be abnormal. Conversely, if the attribute pattern cannot be accurately reconstructed, it is indicated that the node attributes diverges from the pattern observed in most normal nodes. The attribute reconstruction decoder can be represented by:

$$\widehat{\mathbf{X}}_{raw}=\text{GCN}_{Adec}\left(\mathbf{Z}_{raw},\mathbf{A};\mathbf{W}_{Adec}\right), \tag{5}$$

where $\widehat{\mathbf{X}}_{raw}$, $\mathbf{W}_{Adec}$ denote the reconstructed attribute matrix and weight matrix of the attribute decoder.

Topology and attribute regression are two crucial tasks performed by the graph reconstruction module. The reconstruction errors from both types of information are combined, and the objective function can be formulated as follows:

$$\mathcal{L}_{rec} = (1-\alpha)\|\mathbf{A} - \widehat{\mathbf{A}}\|_F^2 + \alpha\|\mathbf{X}_{raw} - \widehat{\mathbf{X}}_{raw}\|_F^2. \tag{6}$$

where α is a parameter employed to balance the reconstruction of topology and attributes.

B. Multi-View Discrimination Module

Contrastive learning has proven to be a successful approach in detecting anomalies on graphs, effectively mitigating the challenges posed by the extreme imbalance between normal and abnormal nodes [19]. In this module, a view-level contrastive-based learning paradigm is designed. By employing this approach, the model efficiently learns the inherent patterns within the graph, thereby augmenting its capability to identify abnormal nodes.

View Generator: Existing node-level contrast approaches relying on subgraph sampling inadequately consider the global structure, resulting in the omission of high-order information. To overcome this limitation, a method for generating positive and negative views is introduced. To enhance the view discrimination ability, inspired by the effectiveness of data augmentation methods in graph self-supervised learning, feature masking is utilized to augment the original features for the positive view. Specifically, the positive view involves randomly masking node attributes by setting the attributes of the masked nodes to zero. This feature masking helps the model to implicitly perform attribute regression, thereby boosting its discriminative capabilities. For the negative view, randomized row transformations are employed on the node features from the original view. It aims to simulate the representation of nodes post-injection with anomalies simply. By completely altering the graph structure, this approach effectively models anomalies in node attributes and topology. To maintain consistency, the feature masking applied to negative views is the same as that used to positive views.

View Generation Encoder: This sub-module aims to extract feature embeddings from the positive and negative views to facilitate the discriminative task within the original features. To maintain consistency, the view generation encoder shares weights with the original view encoder. The view generation encoder can be defined as follows:

$$\mathbf{Z}_{pos} = \mathrm{GCN}_{enc}\left(\mathbf{X}_{pos}, \mathbf{A}; \mathbf{W}_{enc}\right), \tag{7}$$

$$\mathbf{Z}_{neg} = \mathrm{GCN}_{enc}\left(\mathbf{X}_{neg}, \mathbf{A}; \mathbf{W}_{enc}\right), \tag{8}$$

where $\mathbf{X}_{pos}$, $\mathbf{Z}_{pos}$ denote the attribute matrix and node embedding matrix corresponding to the positive views, and $\mathbf{X}_{neg}$ and $\mathbf{Z}_{neg}$ represent the attribute matrix and node embedding matrix for the negative views, respectively.

View Discriminator: The discriminator takes both original and generated view features as inputs. It employs a shared bilinear layer to compute the similarity between these views, which generates a score for discriminating between the positive and negative instances. Facilitating the extraction of intricate relationships and patterns between features, the bilinear layer enables effective discrimination between original and generated views. The similarity between the original view and the positive (negative) view is computed as follows:

$$\mathbf{s}_{pos} = \mathrm{Bilinear}\left(\mathbf{Z}_{raw}, \mathbf{Z}_{pos}; \mathbf{W}_{dis}\right), \tag{9}$$

$$\mathbf{s}_{neg} = \mathrm{Bilinear}\left(\mathbf{Z}_{raw}, \mathbf{Z}_{neg}; \mathbf{W}_{dis}\right), \tag{10}$$

where $\mathbf{s}_{pos}$ represents the similarity between the original view and the positive view, $\mathbf{s}_{neg}$ represents the similarity between the original view and the negative view, and $\mathbf{W}_{dis}$ represents the weight matrix of the discriminator. If a node is deemed normal, the discriminator easily distinguishes between the positives and negatives of the generated view. Conversely, if a node is classified as abnormal, the discriminator struggles to accurately determine the positives and negatives of the generated view.

To promote the discrimination of the learned representation, the binary cross-entropy loss is employed as the objective function:

$$\mathcal{L}_{dis} = -\frac{1}{2N}\sum_{i=1}^{N}\left(y_i \log s_i + (1-y_i)\log\left(1-s_i\right)\right), \tag{11}$$

where y_i represents labels 1 and 0 for positive and negative views respectively, and s_i represents the similarity score between the original view and the generated view of each node v_i. Finally, the loss function of the two modules can be expressed as follows:

$$\mathcal{L} = (1-\beta)\mathcal{L}_{dis} + \beta\mathcal{L}_{rec}. \tag{12}$$

where β is a parameter used to balance the reconstruction module and the discrimination module.

C. Graph Anomaly Scoring Module

After introducing the graph reconstruction module and the multi-view discrimination module, the calculation of anomaly scores for each module is defined.

Graph Reconstruction Score: The graph reconstruction score consists of attribute and topology reconstruction errors. The score $s_{rec,i}$ for each node v_i is calculated as follows:

$$s_{top,i} = \|\mathbf{a}_i - \widehat{\mathbf{a}}_i\|_2^2, \tag{13}$$

$$s_{att,i} = \|\mathbf{x}_i - \widehat{\mathbf{x}}_i\|_2^2, \tag{14}$$

$$s_{rec,i} = (1-\alpha)s_{top,i} + \alpha s_{att,i}, \tag{15}$$

where $\mathbf{a}_i$ denotes the i-th row vector of $\mathbf{A}$, and $\mathbf{x}_i$ denotes the i-th row vector of $\mathbf{X}_{raw}$, $s_{att,i}$ denotes the attribute reconstruction score of v_i and $s_{top,i}$ denotes the topology reconstruction score of v_i. The parameter α is the same as (6) to balance reconstruction module. The higher the reconstruction score is, the more abnormal the node is.

Graph Discrimination Score: Given the nature of similarity scores, normal nodes are expected to have a score close to 1 for positive views and close to 0 for negative views in an ideal case. For each node v_i, the anomaly degree is defined as the difference between positive and negative scores:

$$s_{dis,i} = s_{neg,i} - s_{pos,i}, \tag{16}$$

where $s_{pos,i}$ represents the positive view score of v_i and $s_{neg,i}$ represents the negative view score of v_i.

Graph Anomaly Score: The scores of the reconstruction module and the discrimination module are weighted to produce the composite score a_i, and the formula is as follows:

$$a_i = (1-\beta) s_{dis,i} + \beta s_{rec,i}, \tag{17}$$

where β balances the scores of the two modules, consistent with (12). However, the single-round score calculation does not adequately capture the diverse patterns of anomalies and randomness generated by the negative view. To tackle these challenges, multiple rounds of score averaging are incorporated to obtain the final result. Specifically, the anomaly score calculation for v_i is expressed as:

$$\operatorname{score}(v_i) = \frac{1}{R} \sum_{r=1}^{R} a_i^{(r)}. \tag{18}$$

where $\operatorname{score}(v_i)$ represents the anomaly score of v_i, and R is the number of test rounds. Ideally, increasing the number of rounds for computing the score leads to better stability.

The algorithmic process of the proposed method is presented in Algorithm 1. This algorithm is comprised of two phases: training and testing. During the training phase, the model is trained on two self-supervised tasks: reconstructing the original view and generating positive and negative view contrasts. During the testing phase, iterative rounds of inference computation are employed to derive the final anomaly scores for all nodes.

Algorithm 1: The Algorithmic Process of MDA-GAD.

Input: An attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$; Number of training epoch T, Number of test epoch R.
Output: Anomaly score function $\operatorname{score}(\cdot)$.

1: Initialize trainable parameters $\mathbf{W}_{enc}$, $\mathbf{W}_{Tdec}$, $\mathbf{W}_{Adec}$, and $\mathbf{W}_{dis}$.
2: // Training Stage
3: **for** $t = 1 \rightarrow T$ **do**
4: Generate the positive view $\mathbf{X}_{pos}$ by randomly masking the original feature $\mathbf{X}_{raw}$.
5: Generate the negative view $\mathbf{X}_{neg}$ by randomly masking and row transforming the original feature $\mathbf{X}_{raw}$.
6: Calculate the reconstruction score for all nodes by (13), (14), and (15).
7: Calculate the discrimination score for all nodes by (16).
8: Calculate the overall loss $\mathcal{L}$ by (12).
9: Back propagate and update the trainable parameters $\mathbf{W}_{enc}$, $\mathbf{W}_{Tdec}$, $\mathbf{W}_{Adec}$, and $\mathbf{W}_{dis}$.
10: **end for**
11: // Testing Stage
12: **for** $r = 1 \rightarrow R$ **do**
13: Calculate the composite anomaly score for all nodes by (17).
14: **end for**
15: Calculate the average of the composite scores for multiple rounds by (18).

D. Complexity Analysis

The time complexity of the proposed method comes mainly from its graph reconstruction and multi-view discrimination modules. The graph reconstruction module utilizes a GCN-based encoder and decoder with complexity $\mathcal{O}(LMD + LND^2))$, and the complexity of the matrix reconstruction is $\mathcal{O}(N^2D)$, and the total complexity is $\mathcal{O}(LMD + LND^2 + N^2D)$. The multi-view discrimination module includes a GCN-based encoder and a discriminator with complexities of $\mathcal{O}(LMD + LND^2)$ and $\mathcal{O}(ND^2)$ respectively, summing to $\mathcal{O}(LMD + LND^2 + ND^2)$. Therefore, the total time complexity is $\mathcal{O}((LMD + LND^2 + N^2D)(T + R))$, where L is the number of GCN layers, M is the number of edges, N is the number of nodes, D is the hidden dimension, and T and R are the number of training and inference epochs.

Table II presents a comparison of time complexities between the proposed method and the baseline methods. It is evident that methods employing subgraph sampling exhibit higher complexities, where the complexity of sampling subgraphs is as high as $\mathcal{O}(NP\delta)$. In contrast, the proposed method demonstrates lower complexity while enhancing performance.

TABLE II
COMPARISON OF TIME COMPLEXITY (P AND Q REPRESENT THE NUMBER OF NODES AND EDGES OF THE SUBGRAPH, AND δ IS THE MEAN DEGREE OF NETWORK)

Method	Time Complexity
DOMINANT	$\mathcal{O}((LMD + LND^2 + N^2D)T)$
CoLA	$\mathcal{O}(N(P\delta + LQD + LPD^2)(T + R))$
ANEMONE	$\mathcal{O}(N(P\delta + LQD + LPD^2)(T + R))$
SL-GAD	$\mathcal{O}(N(P\delta + LQD + LPD^2)(T + R))$
Sub-CR	$\mathcal{O}(N^3 + N(P\delta + LQD + LPD^2)(T + R))$
GRADATE	$\mathcal{O}(N(P\delta + LQD + LPD^2)(T + R))$
NLGAD	$\mathcal{O}(N(P\delta + LQD + LPD^2)(T + R))$
ours	$\mathcal{O}((LMD + LND^2 + N^2D)(T + R))$

V. EXPERIMENTAL ANALYSIS

In this section, we describe the datasets used for the experiments and the parameter settings. Then the experimental results are presented, including performance comparison, ablation study, and parameter sensitivity.

A. Dataset Description

Experiments are conducted on six benchmark datasets, consisting of three citation networks, one web citation network and two social networks, as described below.

- *Citation Network:* Cora, Citeseer,[1] and ACM[2] datasets represent citation networks of scientific publications, where nodes represent papers and edges represent citation relationships between papers.

[1]https://github.com/kimiyoung/planetoid
[2]https://github.com/kaize0409/GCN_AnomalyDetection

TABLE III
THE STATISTICS OF THE DATASETS

Dateset	#Nodes	#Edges	#Features	#Anomalies
Cora	2,708	5,429	1,433	150
Citeseer	3,327	4,732	3,703	150
UAI	3,067	28,311	4,973	150
BlogCatalog	5,196	171,743	8,189	300
Flickr	7,575	239,738	12,407	450
ACM	16,484	71,980	8,337	600

- *Web Citation Network:* UAI[3] dataset pertains to a web citation network, where nodes represent pages and edges represent hyperlinks.
- *Social Network:* BlogCatalog and Flickr[4] datasets are social networks, where nodes represent users and edges represent their relationships.

In the mentioned attributed network datasets, inherent anomaly nodes are not present. To facilitate experimentation with the anomaly detection framework, the common practice of introducing anomaly nodes is followed.

Topology Anomaly Injection: The injection of topological anomalies aims to disrupt the relationships between nodes on graphs [41], [42]. This is achieved by randomly selecting t nodes and establishing fully connected links between them. Based on the experience of previous work [19], [20], [21], this process is repeated a total of n times. t is fixed at 15, and n is set to [5, 5, 5, 10, 15, 20] for Cora, Citeseer, UAI, BlogCatalog, Flickr, and ACM, respectively.

Contextual Anomaly Injection: The injection of context exceptions is to disorder the properties of nodes [43]. We randomly select a node v_i, a subset S, find the node v_{far} in the subset S that has the largest distance to the attribute valued by the node v_i, and assign the attribute of v_{far} to v_i. To be consistent with the ratio of topology anomaly injection, contextual anomalies are injected for $t \times n$ nodes.

Finally, an equal proportion of topological and contextual anomalies are introduced into all datasets. By the above injection method, we obtain the perturbed datasets, as shown in Table III.

B. Experimental Settings

In this subsection, we describe the experimental setup, including baseline methods, evaluation metrics, parameter settings, and computing infrastructures.

Baselines: As presented in Table IV, the proposed method considers the strategies more comprehensively than other baseline methods. A brief description of the baseline methods is given as follows:

- *DOMINANT* [16] is an unsupervised anomaly detection framework leveraging deep learning, employing a GCN-based encoder to jointly reconstruct the adjacency and attribute matrix.
- *CoLA* [19] captures anomalous patterns through contrastive self-supervised learning, evaluating consistency between nodes and neighboring subgraphs using a GNN-based encoder.
- *ANEMONE* [20] introduces a multi-scale contrastive learning objective for anomaly detection, where the encoder learns consistency between instances at patch and context levels simultaneously.
- *SL-GAD* [21] is a framework designed to construct various contextual subgraphs centered around target nodes and facilitate anomaly detection through generative attribute regression and multi-view contrast.
- *Sub-CR* [22] is a contrastive learning-based anomaly detection framework that uses graph diffusion to augment the original graph for both inner and outer views.
- *GRADATE* [27] enhances anomaly detection by incorporating subgraph-subgraph and node-node contrasts alongside node-subgraph contrasts within a multi-scale contrast learning framework.
- *NLGAD* [28] is a multi-scale contrastive learning network that incorporates high-confidence nodes into the normality pool, achieving superior results through training based on these nodes.

Evaluation Metric: The anomaly detection performance of all the methods is evaluated using the widely adopted ROC-AUC metric. The ROC curve illustrates the relationship between the true-positive rate and the false-positive rate, and the area under the ROC curve is denoted as AUC. To eliminate the impact of randomness, all experiments are repeated 10 times, and the mean and standard deviation of the results are recorded for comparison.

Parameter Settings: In all experiments, the mask rate p is uniformly set to 0.2. In the basic framework, the encoder employs a two-layer GCN, the attribute decoder employs a two-layer GCN, and the topology decoder employs a one-layer GCN. The hidden dimension is set to 64, and the number of test rounds is fixed to 50. For the learning rate, we use a value of 0.001 for Cora, Citeseer, BlogCatalog, and ACM datasets, 0.0005 for Flickr dataset, and 0.002 for UAI dataset. The number of training epochs is set to 250 for Cora, Citeseer, UAI, Flickr, and ACM datasets, and 300 for BlogCatalog dataset. The α and β parameters perform a grid search between 0.1 and 0.9.

Computing Infrastructures: The experiments for all models are conducted on the PyTorch 1.8.1 platform. The hardware configuration includes an Intel(R) Xeon(R) Silver 4210R CPU operating at a frequency of 2.40 GHz, 56 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory.

C. Experimental Results

In this subsection, we present a performance comparison between the proposed method and seven baseline methods. The ROC curves of all methods on six benchmark datasets are shown in Fig. 3. According to the results, the following conclusions can be drawn:

- Compared to all the baseline methods, the proposed method offers optimal performance across most datasets. As shown in Table V, the proposed method outperforms the current best baseline by 2.81%, 4.06%, 1.49%, 2.12%, 1.23% in

[3]https://github.com/zhumeiqiBUPT/AM-GCN
[4]http://socialcomputing.asu.edu/pages/datasets

TABLE IV
ARCHITECTURE COMPARISON BETWEEN THE PROPOSED METHOD AND BASELINES

Methods	Topology Reconstruction	Attribute Reconstruction	Sampling Free	Contrastive Learning
DOMINANT	✓	✓	✓	×
CoLA	×	×	×	✓
ANEMONE	×	×	×	✓
SL-GAD	×	✓	×	✓
Sub-CR	×	✓	×	✓
GRADATE	×	×	×	✓
NLGAD	×	×	×	✓
Ours	✓	✓	✓	✓

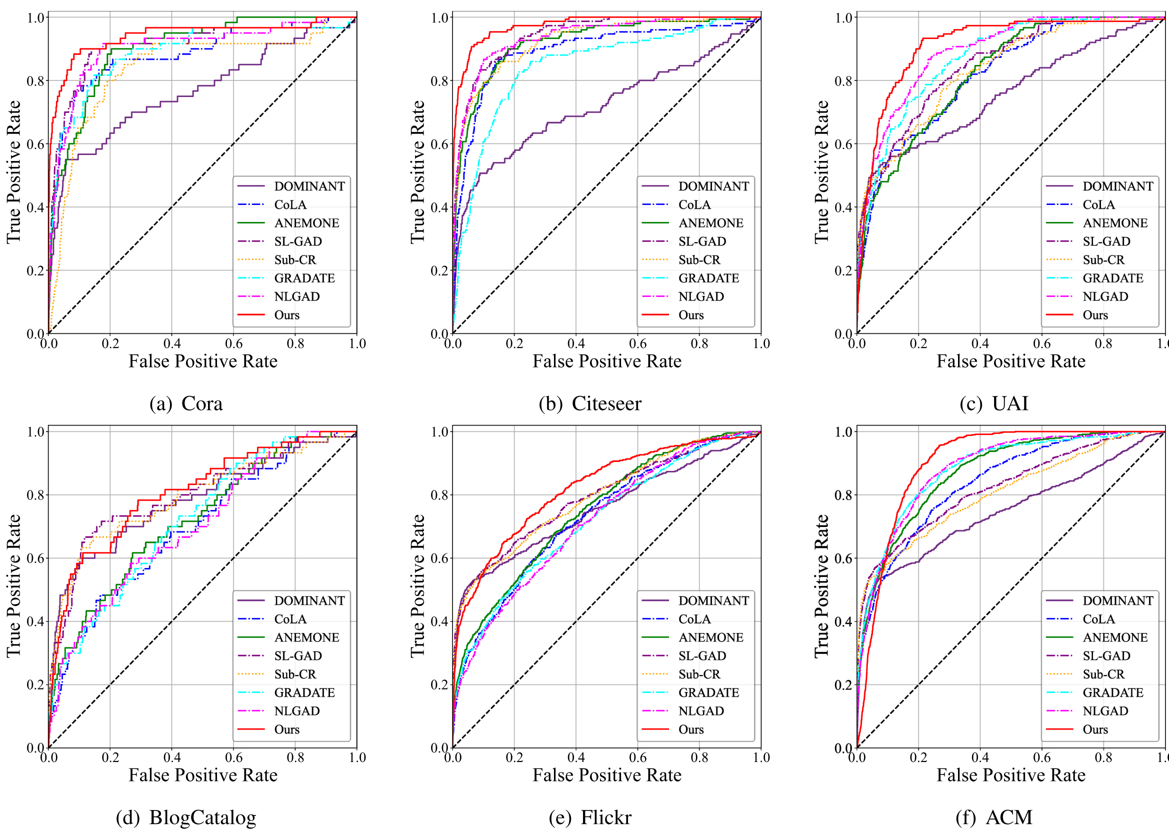
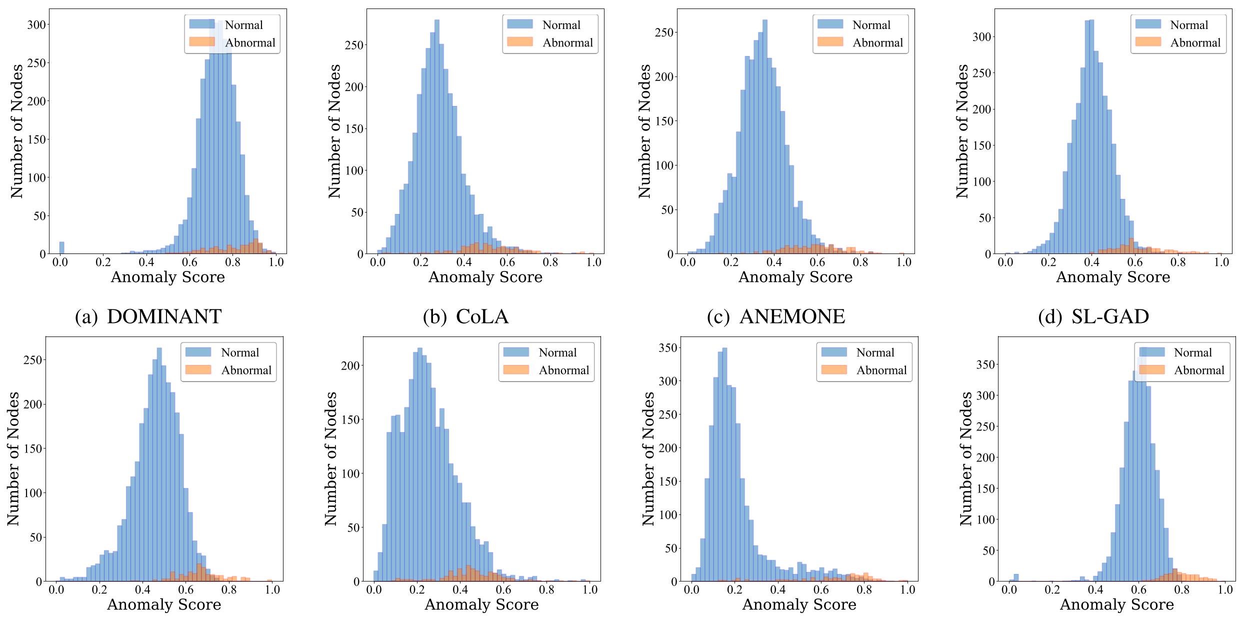


Fig. 3. ROC curves on six benchmark datasets. A larger area under the AUC curve indicates better anomaly detection performance. The black dotted lines refers to the "random line", indicating the performance under random guessing.

TABLE V
COMPARATIVE RESULTS OF GRAPH ANOMALY DETECTION WITH AUC (RUNNING MEANS AND STANDARD DEVIATIONS OF TEN TRIALS)

Methods	Cora	Citeseer	UAI	BlogCatalog	Flickr	ACM
DOMINANT	0.7604 (0.0035)	0.7195 (0.0015)	0.7553 (0.0003)	0.7821 (0.0000)	0.7442 (0.0001)	0.7479 (0.0000)
CoLA	0.8884 (0.0154)	0.8960 (0.0226)	0.8104 (0.0082)	0.7031 (0.0115)	0.7348 (0.0042)	0.8263 (0.0129)
ANEMONE	0.9037 (0.0080)	0.9163 (0.0182)	0.8343 (0.0198)	0.7475 (0.0092)	0.7566 (0.0121)	0.8687 (0.0100)
SL-GAD	0.9159 (0.0085)	0.9288 (0.0138)	0.8452 (0.0040)	**0.8044 (0.0104)**	0.7940 (0.0038)	0.8141 (0.0045)
Sub-CR	0.8993 (0.0072)	0.9198 (0.0098)	0.8355 (0.0028)	0.7930 (0.0052)	0.7973 (0.0031)	0.7896 (0.0048)
GRADATE	0.9056 (0.0071)	0.8871 (0.0111)	0.8873 (0.0112)	0.7095 (0.0243)	0.7103 (0.0093)	0.8853 (0.0113)
NLGAD	0.9075 (0.0109)	0.9304 (0.0150)	0.8920 (0.0049)	0.6995 (0.0092)	0.7081 (0.0076)	0.8904 (0.0052)
Proposed	**0.9440 (0.0032)**	**0.9710 (0.0054)**	**0.9069 (0.0067)**	0.8034 (0.0028)	**0.8185 (0.0062)**	**0.9027 (0.0005)**

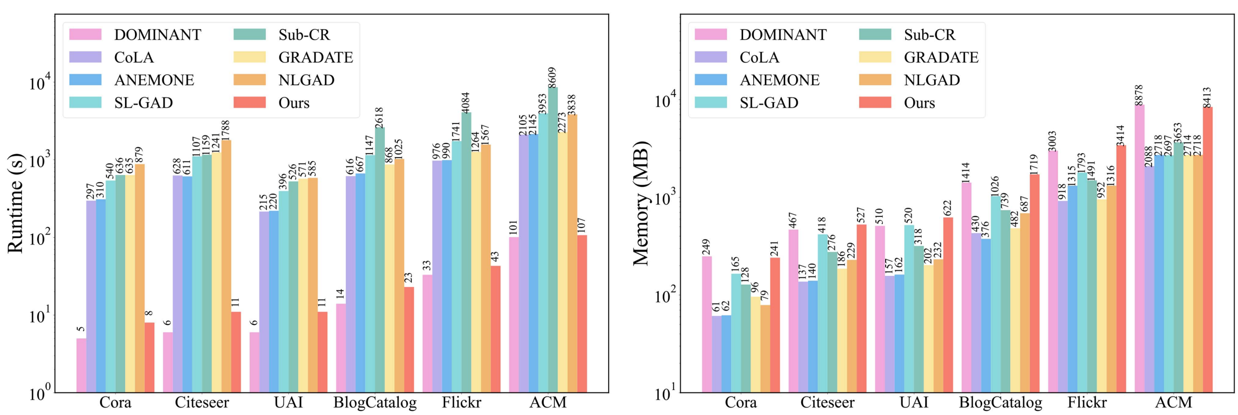
The best and second best results in each column are highlighted in bold and underlined, respectively.



(a) DOMINANT (b) CoLA (c) ANEMONE (d) SL-GAD

(e) Sub-CR (f) GRADATE (g) NLGAD (h) Ours

Fig. 4. Visualization of the anomaly score distribution for eight methods on the Citeseer dataset. The scores are normalized to a range of 0 to 1.



(a) Runtime (b) Maximum memory usage

Fig. 5. Runtime and maximum memory usage comparison between the proposed method and baselines on six benchmark datasets.

Cora, Citeseer, UAI, Flickr, and ACM respectively. Fig. 4 shows the distribution of anomaly scores for different comparison methods on Citeseer. It is noticed that the anomaly score distribution for the proposed method is more centralized, and the distinction in the score distribution between normal and abnormal nodes is more prominent.

- The experiments comparing the runtime and maximum memory usage of all methods are shown in Fig. 5. The proposed method excels in time efficiency, second only to DOMINANT. Compared to the suboptimal model, the proposed method has a runtime 53 times faster than SL-GAD and 49 times faster than NLGAD. Besides, the proposed method has a higher memory usage compared to baseline methods, mainly because it utilizes the full graph input, whereas most methods employ batch training. Although batch training reduces memory consumption, it significantly increases the time overhead due to its inherent complexity.
- Compared to models that solely rely on contrastive self-supervised learning methods like CoLA, ANEMONE, GRADATE, and NLGAD, and models that exclusively utilize reconstruction methods like DOMINANT, the proposed method demonstrates superior anomaly detection and generalization capabilities. A pivotal factor contributing to this advantage lies in the integration of two self-supervised methods, which harnesses a more diverse set of supervised signals.
- Compared to hybrid learning paradigms like SL-GAD and Sub-CR, the proposed method also exhibits strong performance. Research has demonstrated that hybrid

TABLE VI
RESULTS OF ABLATION EXPERIMENTS (RUNNING MEANS AND STANDARD DEVIATIONS OF TEN TRIALS)

	Cora	Citeseer	UAI	BlogCatalog	Flickr	ACM
w/o Rec	0.4950 (0.0361)	0.4863 (0.0276)	0.3962 (0.0059)	0.4592 (0.0041)	0.4188 (0.0038)	0.4958 (0.0230)
w/o Dis	0.9235 (0.0019)	0.9449 (0.0018)	0.7992 (0.0010)	0.7695 (0.0002)	0.7375 (0.0001)	0.8506 (0.0000)
w/o Pos	0.9371 (0.0053)	0.9577 (0.0069)	0.8963 (0.0096)	0.8019 (0.0016)	0.7724 (0.0079)	0.8823 (0.0007)
w/o Neg	<u>0.9407 (0.0055)</u>	<u>0.9708 (0.0072)</u>	<u>0.9045 (0.0060)</u>	<u>0.8022 (0.0032)</u>	<u>0.8128 (0.0037)</u>	<u>0.8861 (0.0004)</u>
Ours	**0.9440 (0.0032)**	**0.9710 (0.0054)**	**0.9069 (0.0067)**	**0.8034 (0.0028)**	**0.8185 (0.0062)**	**0.9027 (0.0005)**

The best and second best results in each column are highlighted in bold and underlined, respectively.

learning approaches can produce both suboptimal and optimal outcomes, underscoring the effectiveness of this strategy. Nonetheless, SL-GAD and Sub-CR rely on subgraph sampling, limiting their ability to estimate attributes solely based on sampled neighbors. This constraint may lead to information loss and an inability to fully exploit the potential of the reconstruction module. In contrast, the proposed method, utilizing complete graph training, avoids these drawbacks and maintains the effectiveness of the reconstruction module.

- The proposed method has exhibited exceptional performance in citation networks. However, there is still potential for further improvements to augment its effectiveness in the context of social networks. This is because social networks typically have a significantly higher average node degree compared to citation networks, resulting in more node relationships that pose a greater challenge for graph anomaly detection.

D. Ablation Study

In this subsection, we perform an ablation study to assess the effectiveness of the graph reconstruction module and the multi-view discrimination module. Specifically, four scenarios are evaluated: w/o Res, w/o Dis, w/o Pos, and w/o Neg. In the w/o Res scenario, the graph reconstruction module is excluded, leading the model to rely solely on the multi-view discrimination module during training. In the w/o Dis scenario, the multi-view discrimination module is entirely removed, resulting in no discriminant scores being utilized in the training process. In the w/o Pos scenario, only the scores of the generated negative views are used as discriminant scores, while in the w/o Neg scenario, only the scores of the generated positive views are employed. The conclusions drawn from the results in Table VI are as follows:

- After removing the graph reconstruction module, the performance of using only the multi-view discrimination module significantly declines. This is likely because, without joint learning, the original views containing anomalies cannot be effectively distinguished at the view level by contrastive learning alone. In other words, the absence of the graph reconstruction module prevents the model from capturing and reconstructing node topology and node attributes of the graph, which consequently affects the discriminative ability of the multi-view discrimination module.
- Although the graph reconstruction module has shown performance improvements on specific datasets like Cora and Citeseer, its adaptability to a broader range of scenarios remains limited. The success of the proposed method across most datasets underlines the effectiveness and importance of integrating the multi-view discrimination module. This component enhances the its capacity to capture and leverage information from diverse perspectives, leading to an overall performance enhancement. Therefore, the joint learning of both modules is crucial for achieving robust anomaly detection.
- Across all datasets, the multi-view discrimination module that utilizes only positive view scores consistently has higher AUC values compared to modules that rely solely on negative view scores. Nevertheless, the failure of the positive view-only scoring approach to take advantage of the variable anomaly patterns present in the negative views leads to sub-optimal results. Similarly, due to the uncertainty associated with these patterns, relying solely on the negative view score may lead to model instability. Therefore, a more effective approach is to combine the two views, leveraging their respective strengths to achieve better performance.

E. Parameters Sensitivity

In this subsection, we conduct experiments to assess the impacts of various hyperparameters. These hyperparameters include the balancing parameters α and β, the mask rate p, the hidden dimension D, and the number of test rounds R.

Influence of the balancing parameters: Fig. 6 shows a demonstration of the effect of the balancing parameter under different datasets. The balancing parameter α is responsible for controlling the weighting between attribute reconstruction and topology reconstruction. After analyzing the obtained parameters, it is evident that attribute reconstruction is typically assigned to a higher weight proportion in the proposed method. Similarly, the balance parameter β governs the weights of the reconstruction module and the view discrimination module. For Cora and Citeseer datasets, both the reconstruction module and the discrimination module play equally important roles. On the contrary, for UAI and Flickr datasets, the discrimination module proves to be more advantageous, while for BlogCatalog and ACM datasets, the reconstruction module demonstrates greater utility.

Influence of the mask rate: To further investigate the impact of mask rate size on the generated views across different datasets, a detailed analysis is conducted. Fig. 7(a) shows the AUC values corresponding to different mask rates. It indicates a slight increase in the AUC value as the mask rate reaches 0.2. However,

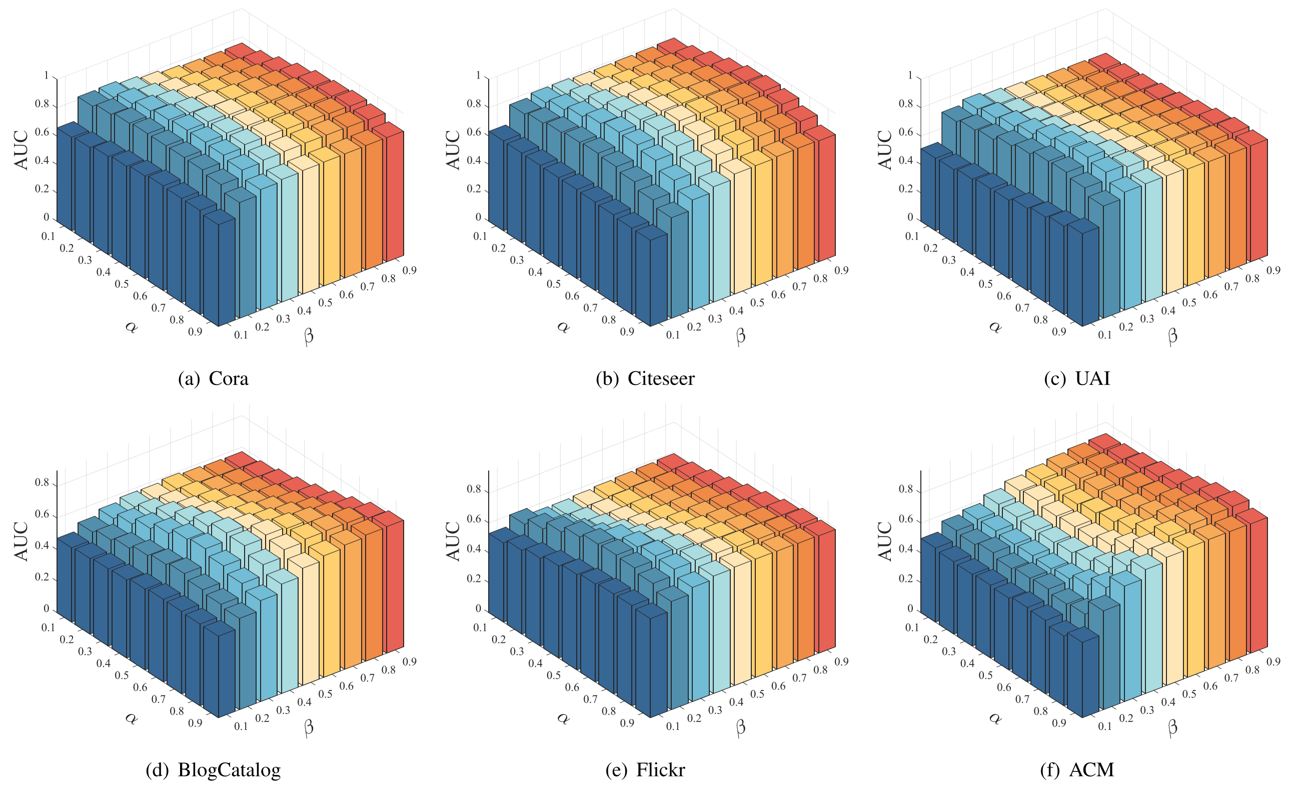


Fig. 6. The experimental results for the investigation of α and β parameters. These subfigures show the effect of the α and β parameters on the AUC value.

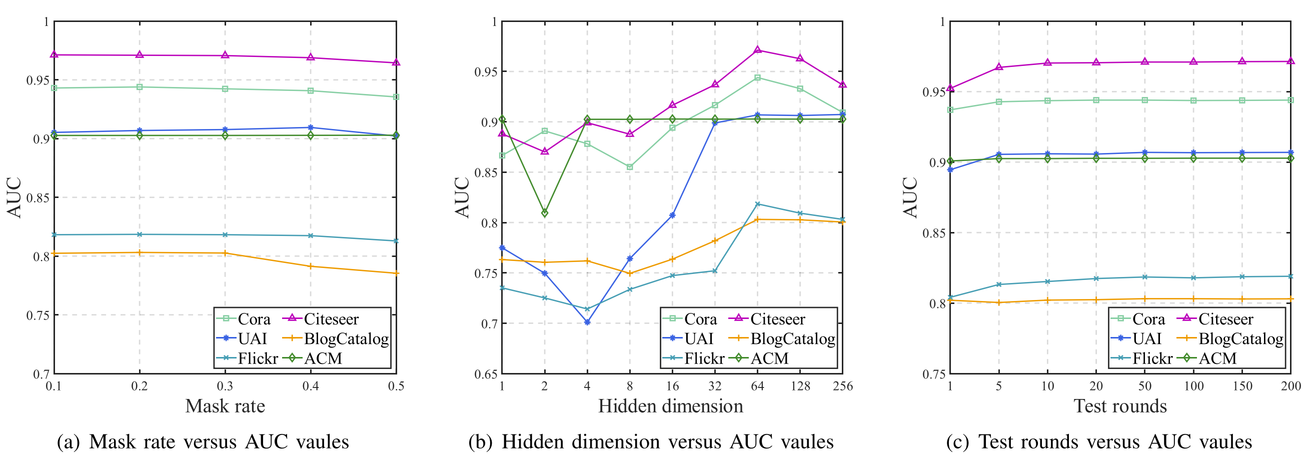


Fig. 7. The experimental results of the parametric study are presented. These subfigures show the effectiveness of the masking rate, embedding dimension, and number of test rounds on the AUC values, respectively.

it starts to decrease after exceeding this threshold. The decline observed in this case can be attributed to an excessively high mask rate. High rate disrupts the inherent pattern of the original view, which hampers the effective differentiation between positive and negative views. Based on the analysis, a mask rate of 0.2 is a suitable choice for generating views.

Influence of the hidden dimension: In this experiment, we investigate the effect of hidden dimensions on the framework's performance, as illustrated in Fig. 7(b). Notably, an enhancement in anomaly detection performance is observed as the hidden dimension increases from 1 to 64. Nonetheless, surpassing this threshold results in a decline of performance. This decline may be due to the increased risk of overfitting associated with the increase in model parameters. Therefore, it is concluded that a 64-dimensional hidden embedding suffices to provide adequate information for subsequent anomaly detection tasks across most datasets.

Influence of the number of test rounds: In this experiment, we further explore the influence of test rounds on the model across different R values. Fig. 7(c) illustrates the variation of AUC values. As the R value increases, the AUC stabilizes and shows some improvement. Based on the experimental findings, only 50 tests are necessary to stabilize the results. This enhancement can be attributed to the increased number of rounds, which helps mitigate the impact of randomness generated in the multi-view discrimination module and allows for the capture of a broader range of negative pattern information. However, excessively high numbers of rounds can lead to an escalation in the computational overhead of the model. Nevertheless, compared to methods relying on subgraph sampling, the proposed method exhibits superior performance with fewer rounds.

VI. Conclusion

In this paper, we proposed a novel hybrid method via multi-view discriminative awareness learning for graph anomaly detection. The anomaly detection task was completed through two distinct self-supervised modules, namely the graph reconstruction module and the multi-view discrimination module. The graph reconstruction module effectively identified the degree of anomalies of a node by leveraging reconstruction errors. The multi-view discrimination module provided more diverse supervised signals by generating positive and negative views for view-level contrast. The experimental results on six benchmark datasets demonstrated that the proposed method outperformed existing state-of-the-art models in terms of AUC when solving graph anomaly detection problems. Despite the superior performance gained by the proposed method, a drawback is the intensive memory utilization, particularly when processing large-scale datasets. Consequently, our future work will concentrate on the memory optimization to address this concern.

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